

PROJECT BUTTON HR–V0 ELECTRICAL V3 CONNECTED CANDIDATE

Direct dual–channel E–stop inputs, watchdog–gated SR1 supply, separate RESET/ARM, redundant actuator interruption.

01 External listed sources and DC boundaries

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02 Dual–channel E–stop and RESET eligibility

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03 Distinct ARM and watchdog–gated SR1 supply

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04 Contactor coils, mirror contacts and EDM

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11 Watchdog PCB external connectors

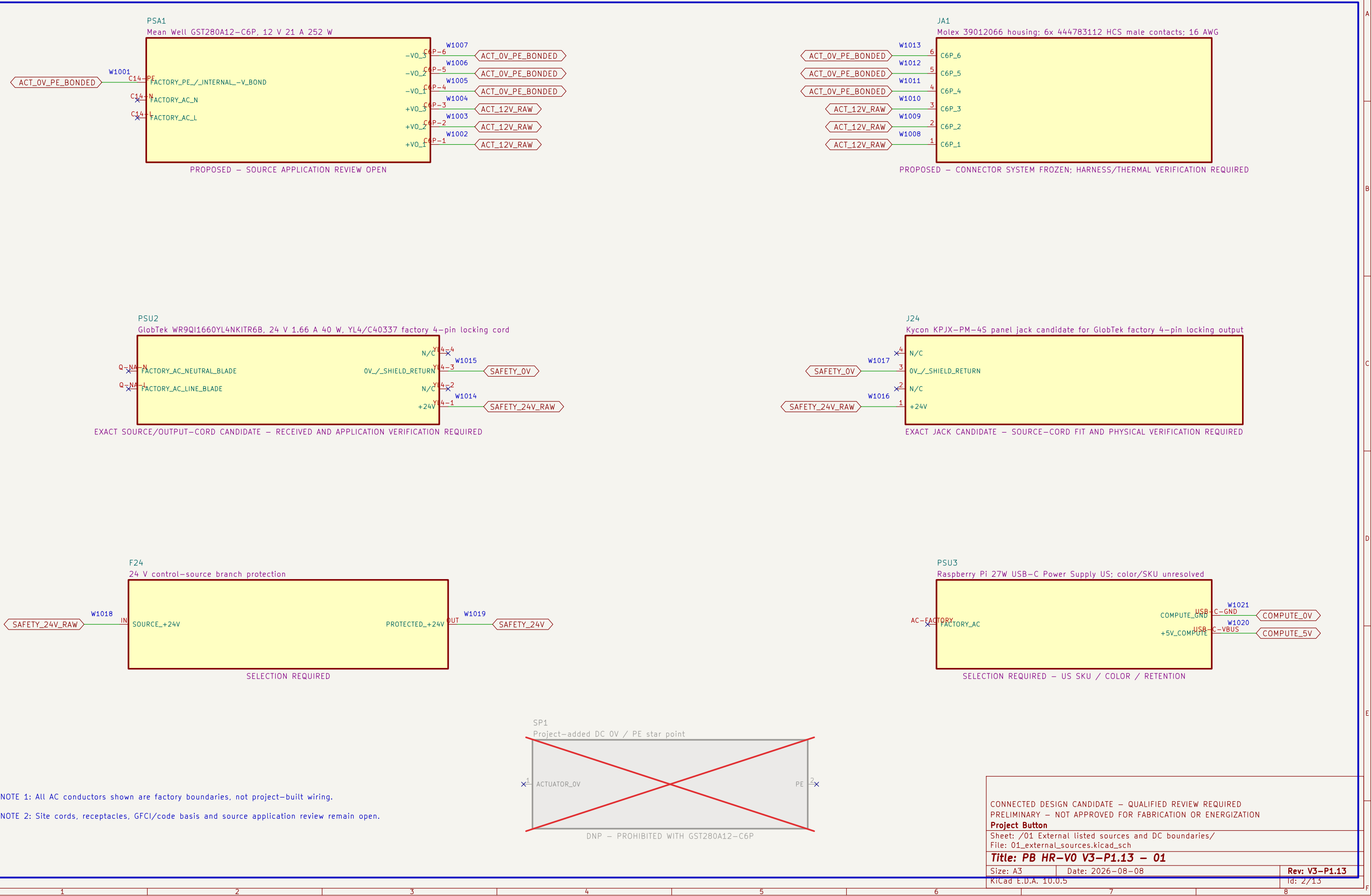
File: 11\_watchdog\_pcb\_connectors.kicad\_sch

12 Watchdog PCB test access

File: 12\_watchdog\_pcb\_test\_access.kicad\_sch

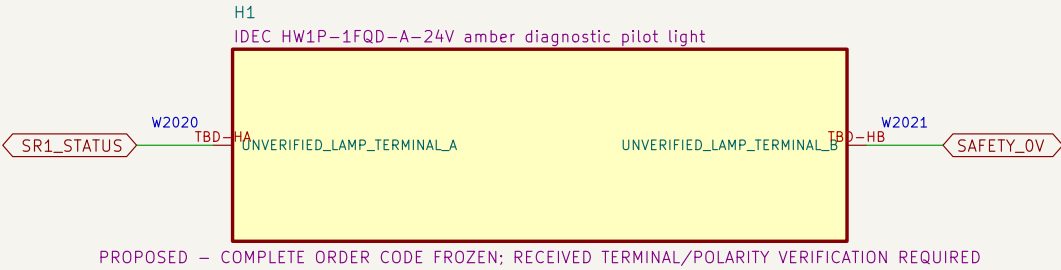
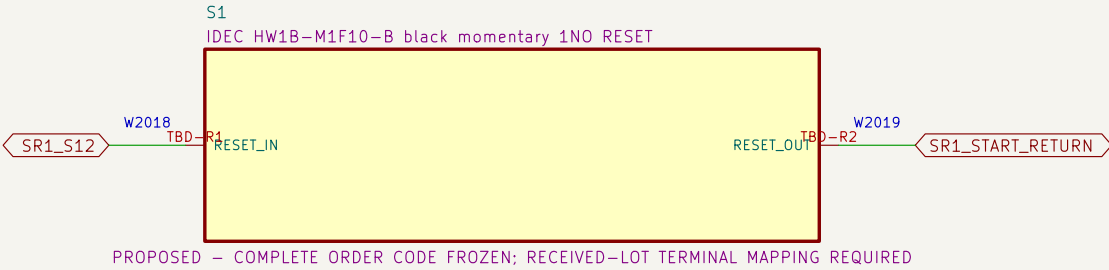
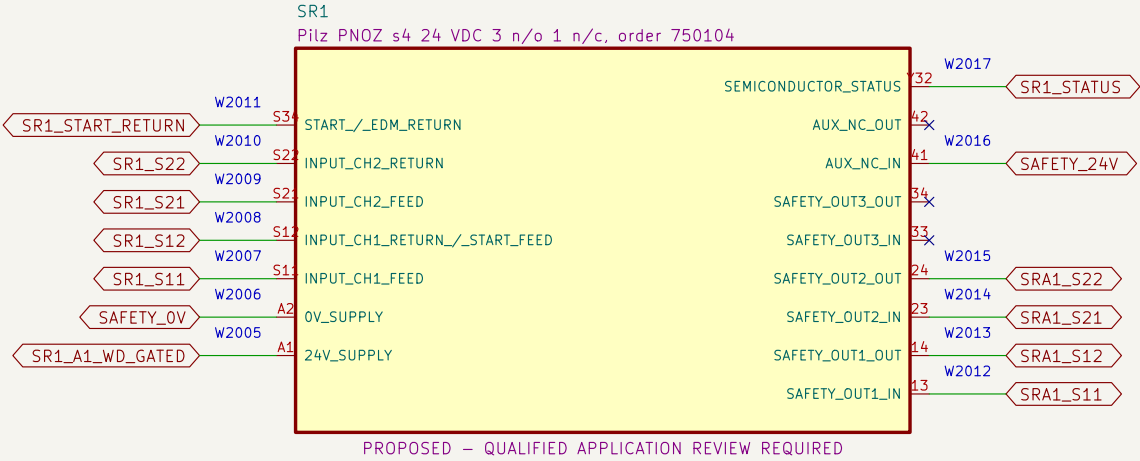
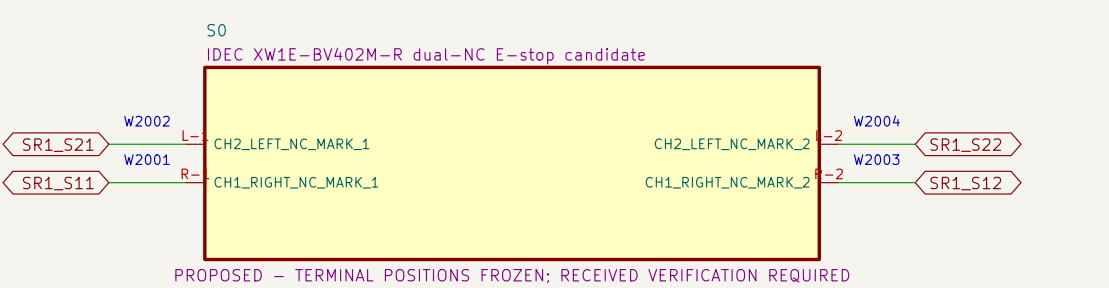
01 External listed sources and DC boundaries

Factory-sealed adapters eliminate project-built mains wiring.



02 Dual-channel E-stop and RESET eligibility

Each SR1 input return contains only one S0 NC contact; RESET cannot energize K1/K2.



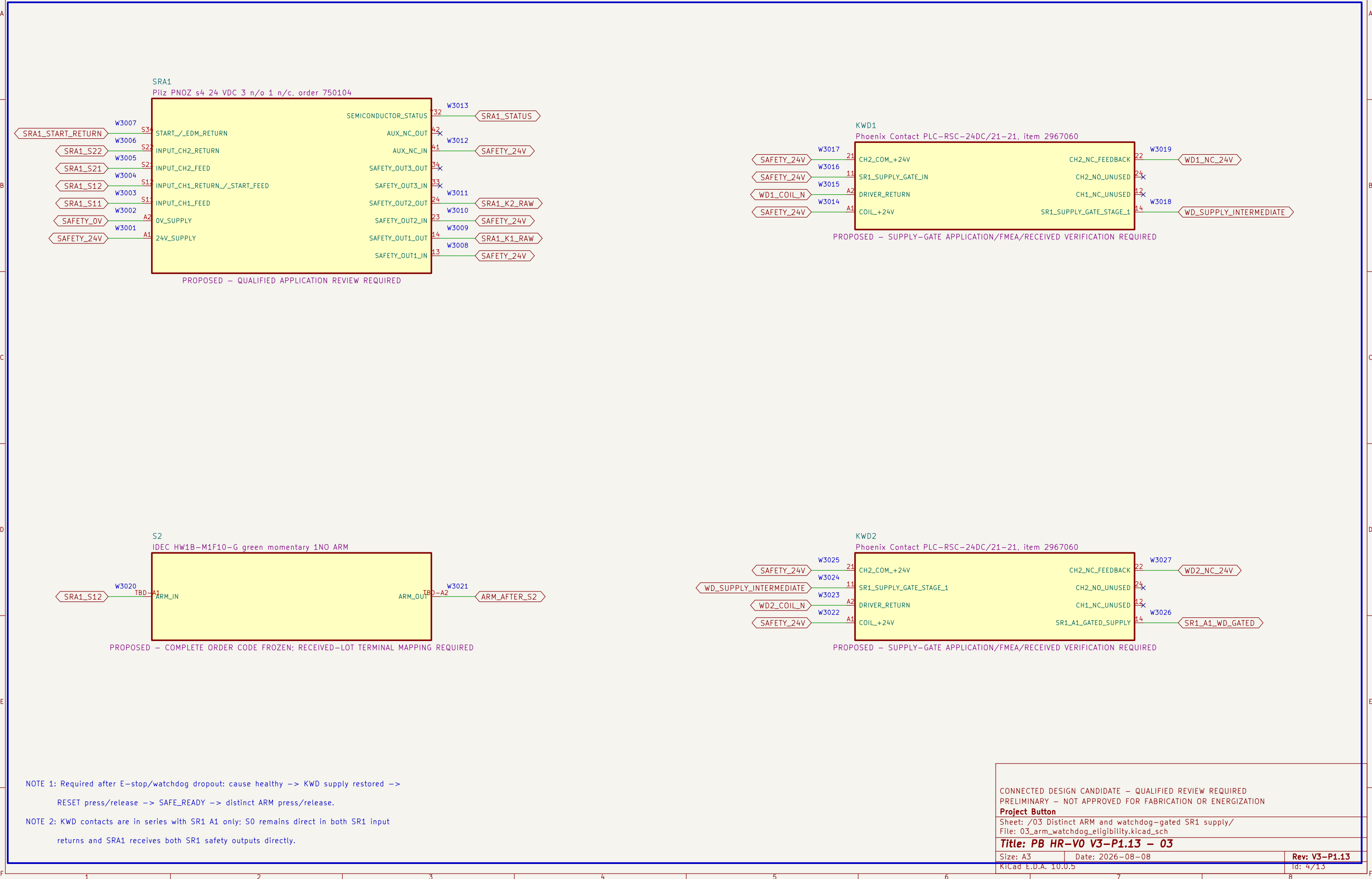
NOTE 1: S0 directly completes both SR1 input returns; ordinary KWD terminals are absent from both loops.

NOTE 2: Heartbeat loss removes SR1 A1 through the separate two-contact supply gate; recovery alone cannot restore the monitored RESET stage.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED  
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION  
**Project Button**  
Sheet: /02 Dual-channel E-stop and RESET eligibility/  
File: 02\_estop\_eligibility.kicad\_sch

03 Distinct ARM and watchdog-gated SR1 supply

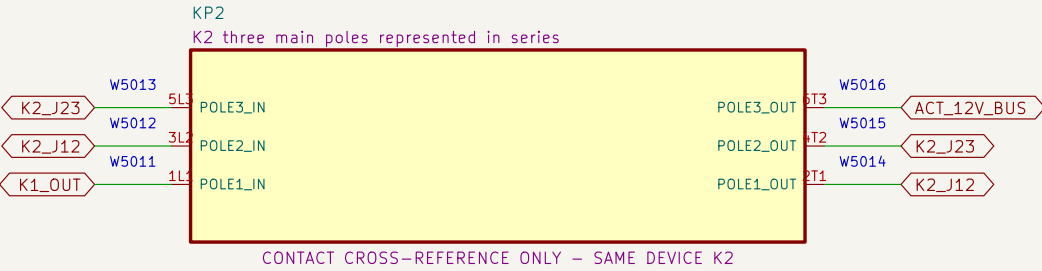
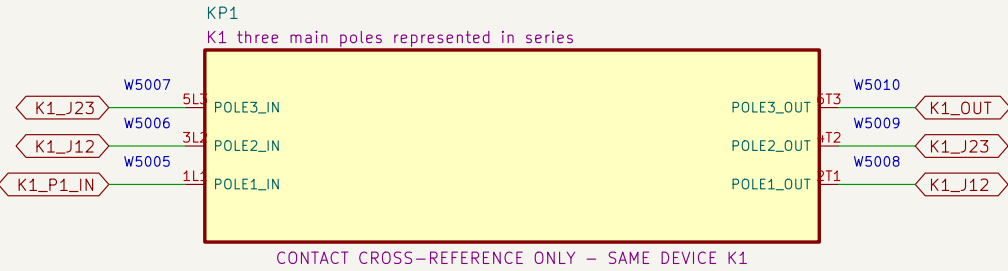
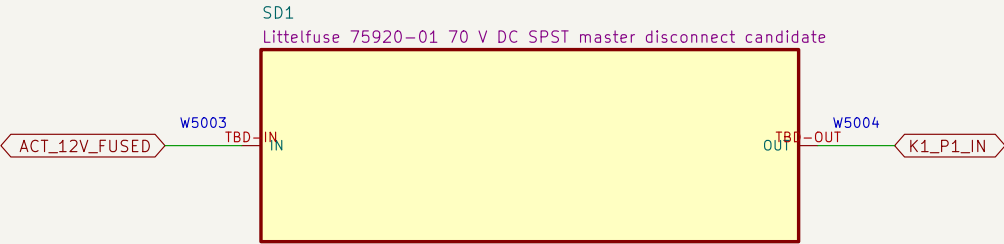
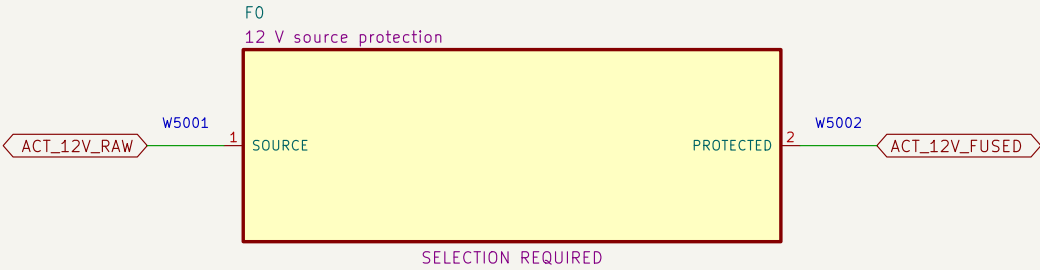
Two ordinary KWD contacts gate SR1 A1; SRA1 still requires SR1 outputs, EDM proof and a new ARM action.





05 Redundant actuator–power interruption

Source protection, service disconnect and all three poles of K1 then K2 are represented in series.



NOTE 1: Series jumpers: K1 2T1–>3L2, 4T2–>5L3, 6T3–>K2 1L1; repeat through K2 to ACT\_12V\_BUS.

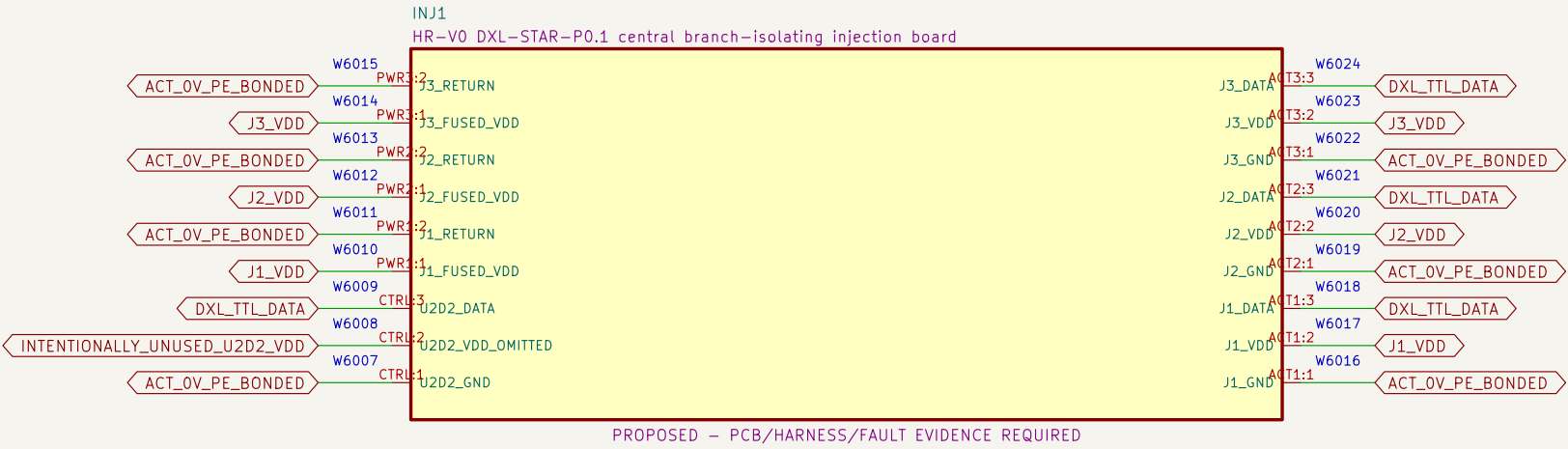
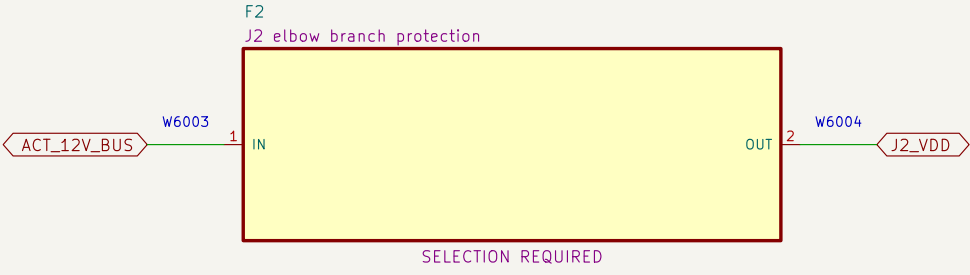
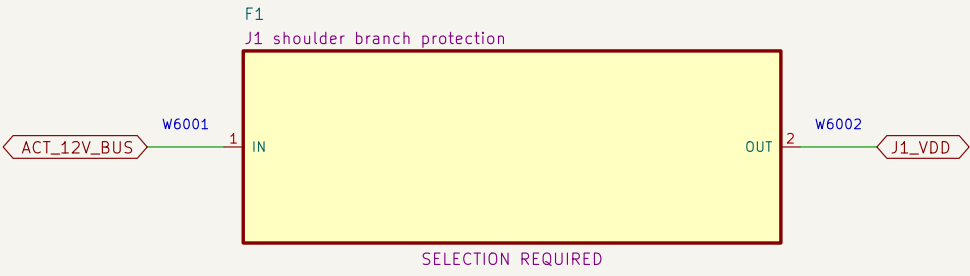
NOTE 2: No fuse, disconnect, conductor, connector or contactor application is released without fault–current evidence.

|                                                                                               |                  |               |
|-----------------------------------------------------------------------------------------------|------------------|---------------|
| CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED                                        |                  |               |
| PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION                                    |                  |               |
| Project Button                                                                                |                  |               |
| Sheet: /05 Redundant actuator–power interruption/<br>File: 05_actuator_interruption.kicad_sch |                  |               |
| Title: PB HR–V0 V3–P1.13 – 05                                                                 |                  |               |
| Size: A3                                                                                      | Date: 2026–08–08 | Rev: V3–P1.13 |
| KiCad E.D.A. 10.0.5                                                                           |                  | Id: 6/13      |



06 Protected actuator branches and central DYNAMIXEL star injection

Each actuator has a separate protected VDD branch; U2D2 pin 2 and inter-actuator VDD paths are omitted.



NOTE 1: INJ1 is a fixed central star; U2D2 pin 2 is unrouted and J1/J2/J3 positive rails never join.

NOTE 2: Exact actuator cables, branch returns, enclosure, continuity, isolation, signal-integrity, pull and no-backfeed tests remain release gates.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED  
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION

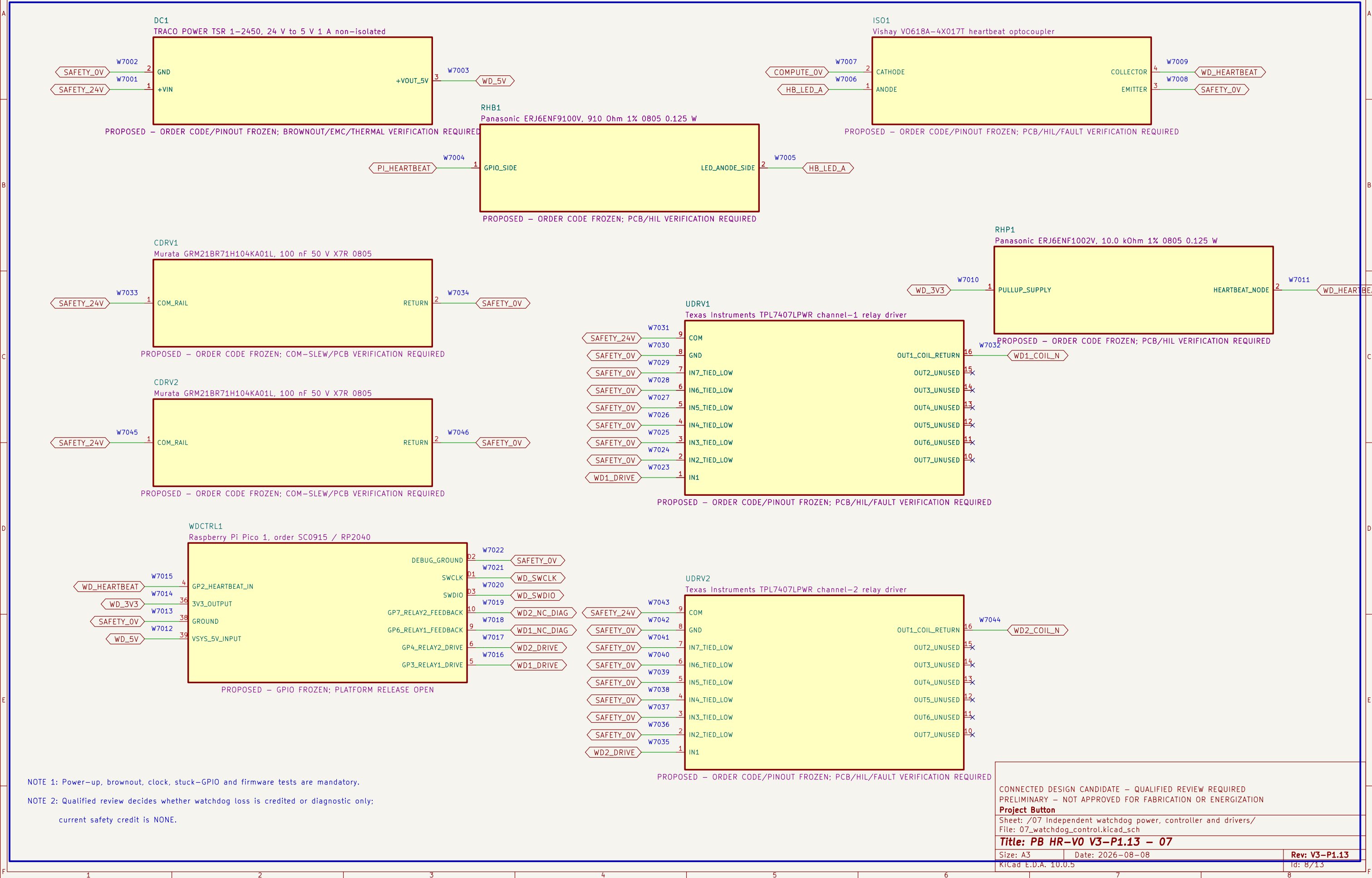
**Project Button**  
Sheet: /06 Protected actuator branches and central DYNAMIXEL star injection/  
File: 06\_branches\_and\_injection.kicad\_sch

**Title: PB HR-V0 V3-P1.13 – 06**

|                     |                  |               |
|---------------------|------------------|---------------|
| Size: A3            | Date: 2026-08-08 | Rev: V3-P1.13 |
| KiCad E.D.A. 10.0.5 |                  | Id: 7/13      |

07 Independent watchdog power, controller and drivers

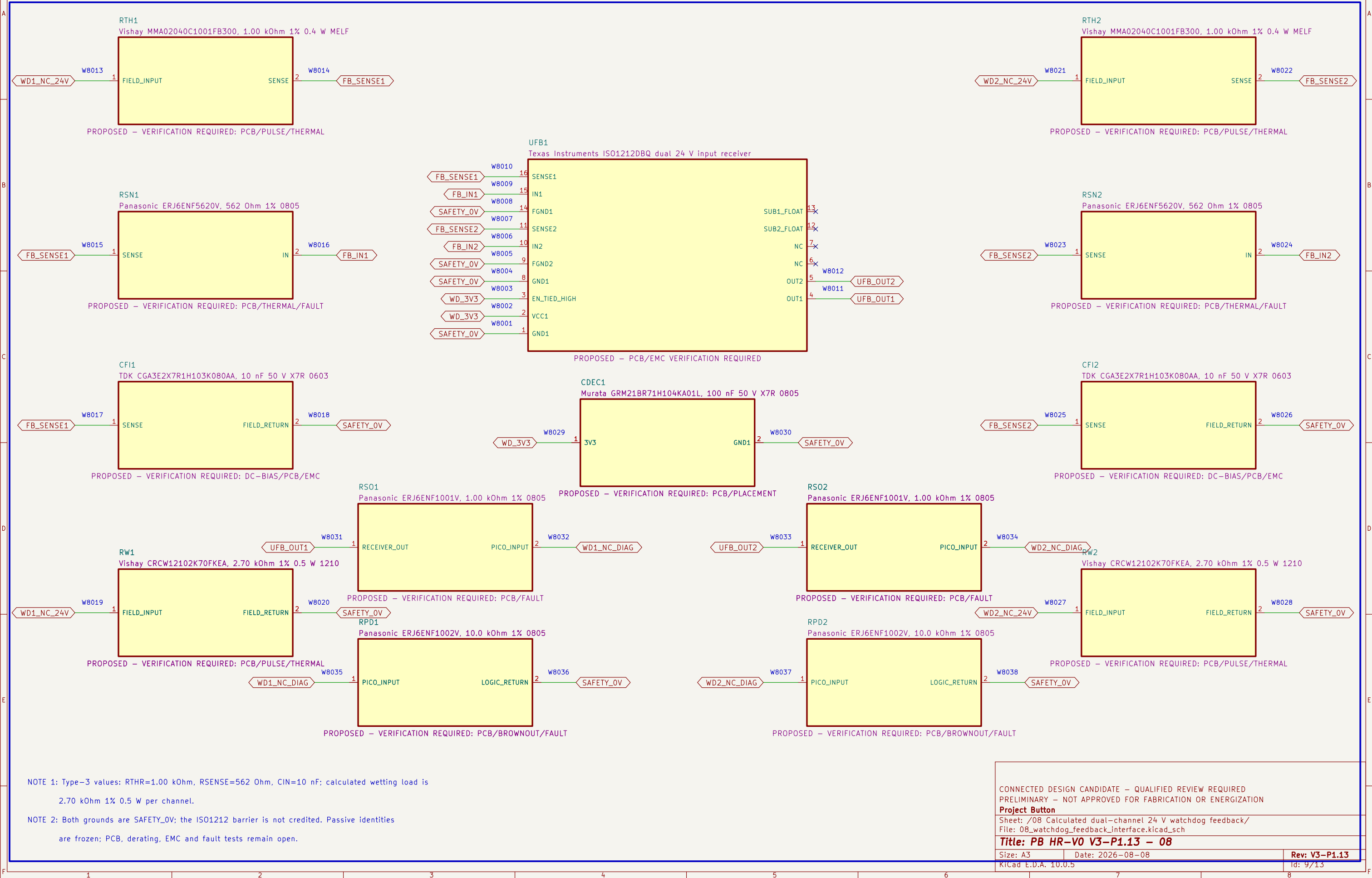
The ordinary watchdog controller and drivers receive no safety–integrity credit by assertion.





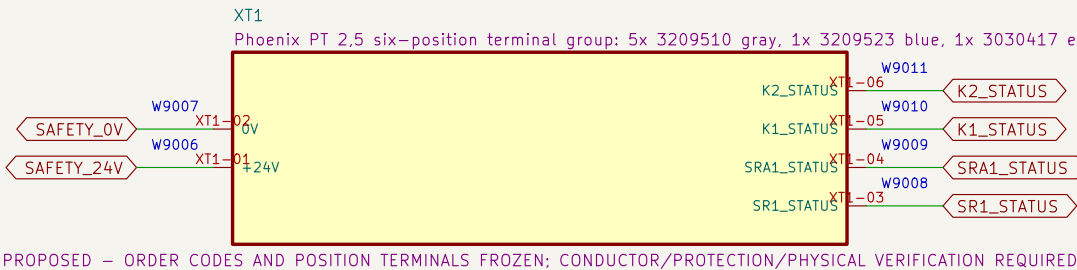
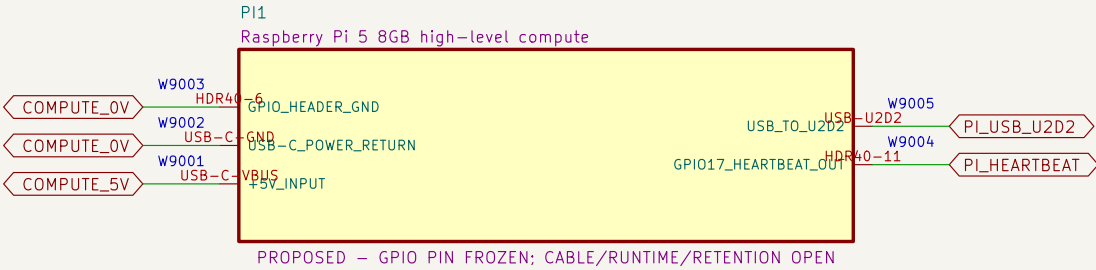
08 Calculated dual-channel 24 V watchdog feedback

ISO1212DBQ converts both relay NC diagnostics to default-low 3.3 V logic; no isolation or safety credit is claimed.



09 Compute and control terminals

High-level compute and diagnostic wiring have no authority to bypass or restore the safety chain.



NOTE 1: No installed debug connector exists; watchdog programming uses only TP15/TP16/TP2 under a future controlled unpowered-fixture procedure.

NOTE 2: Debug connection, halt, flash and disconnect must not enable outputs, back-power the board or bypass a protective function.

NOTE 3: Heartbeat cable, GPIO runtime, timing, terminal family, markers, conductor range, torque and enclosure layout remain selection required.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED  
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION  
**Project Button**

Sheet: /09 Compute and control terminals/  
File: 09\_compute\_and\_control\_terminals.kicad\_sch

**Title: PB HR-V0 V3-P1.13 – 09**

Size: A3 Date: 2026-08-08

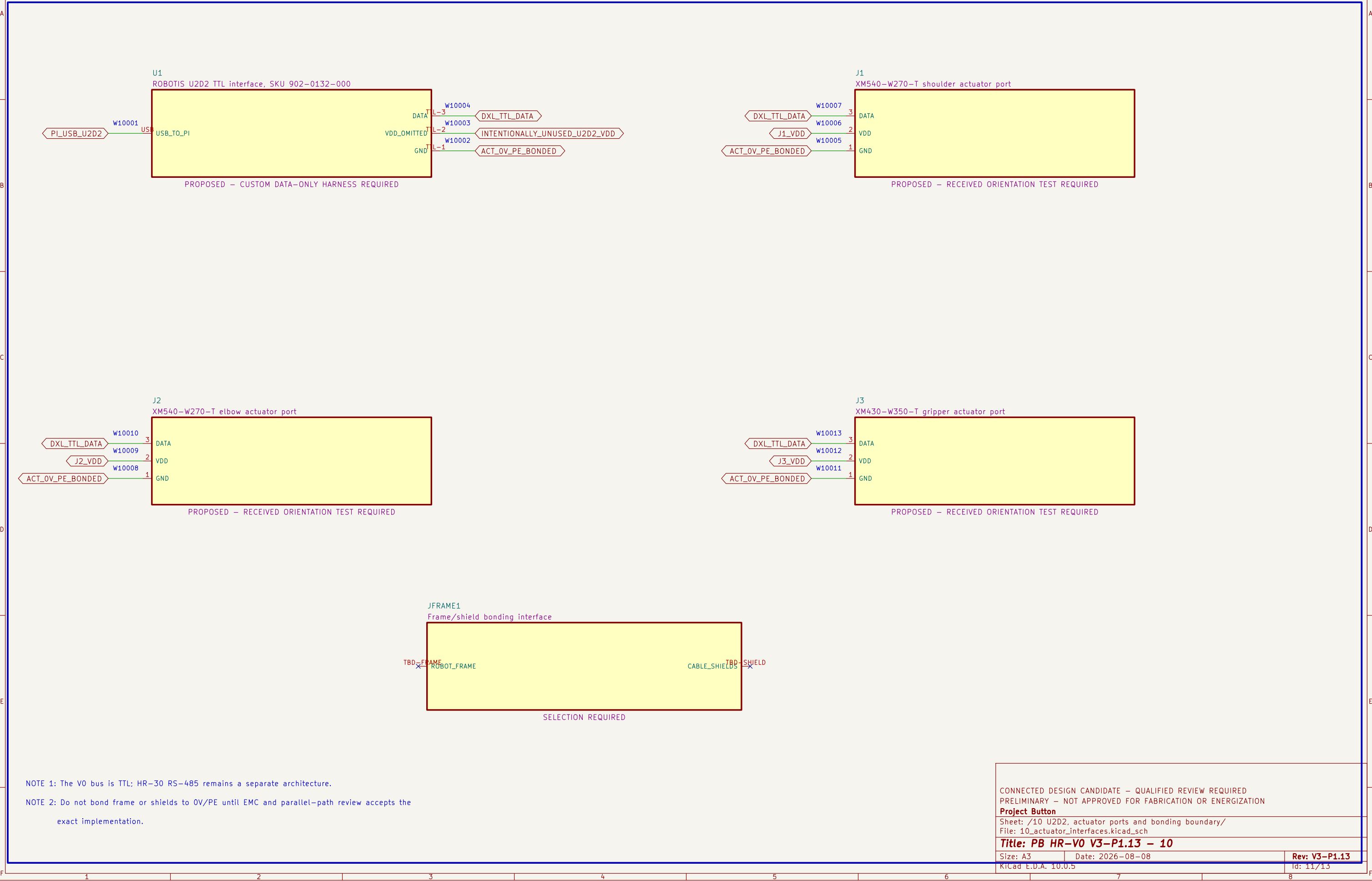
Rev: V3-P1.13

KiCad E.D.A. 10.0.5

Id: 10/13

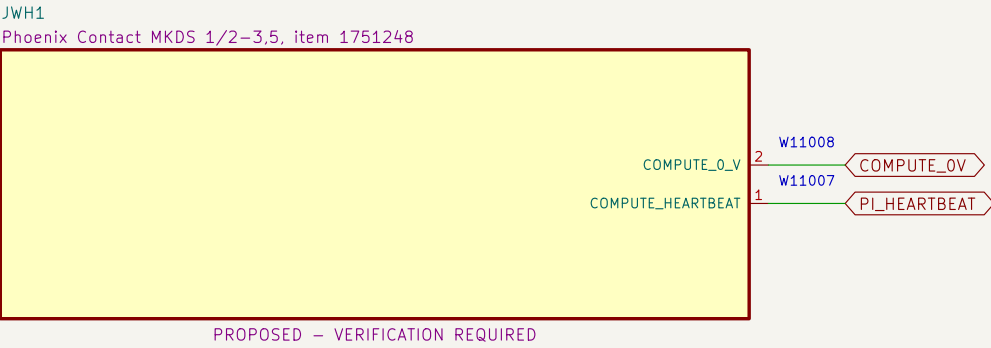
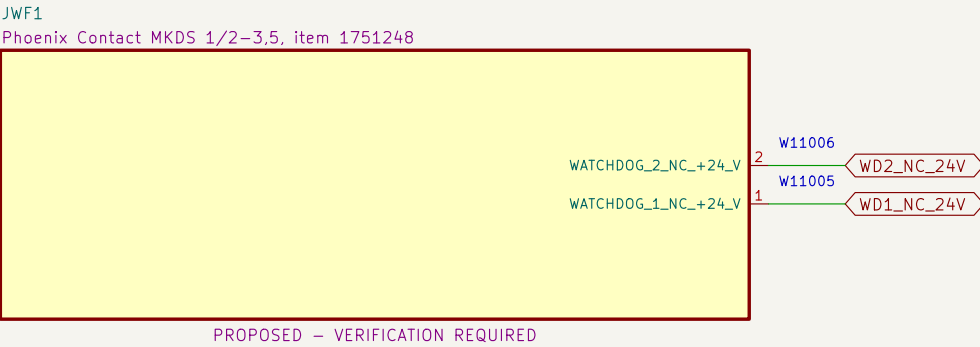
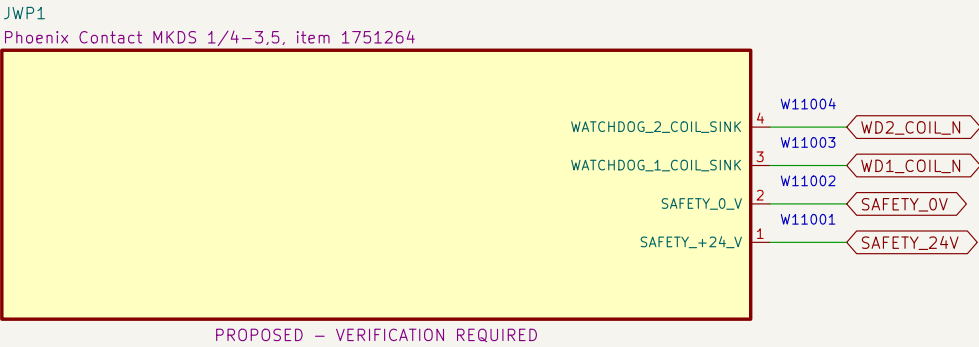
10 U2D2, actuator ports and bonding boundary

The U2D2 cable carries DATA and GND only; protected VDD is injected at each actuator.



11 Watchdog PCB external connectors

Exact PCB terminal–block candidates and project pin allocations define the board boundary; harness release remains open.



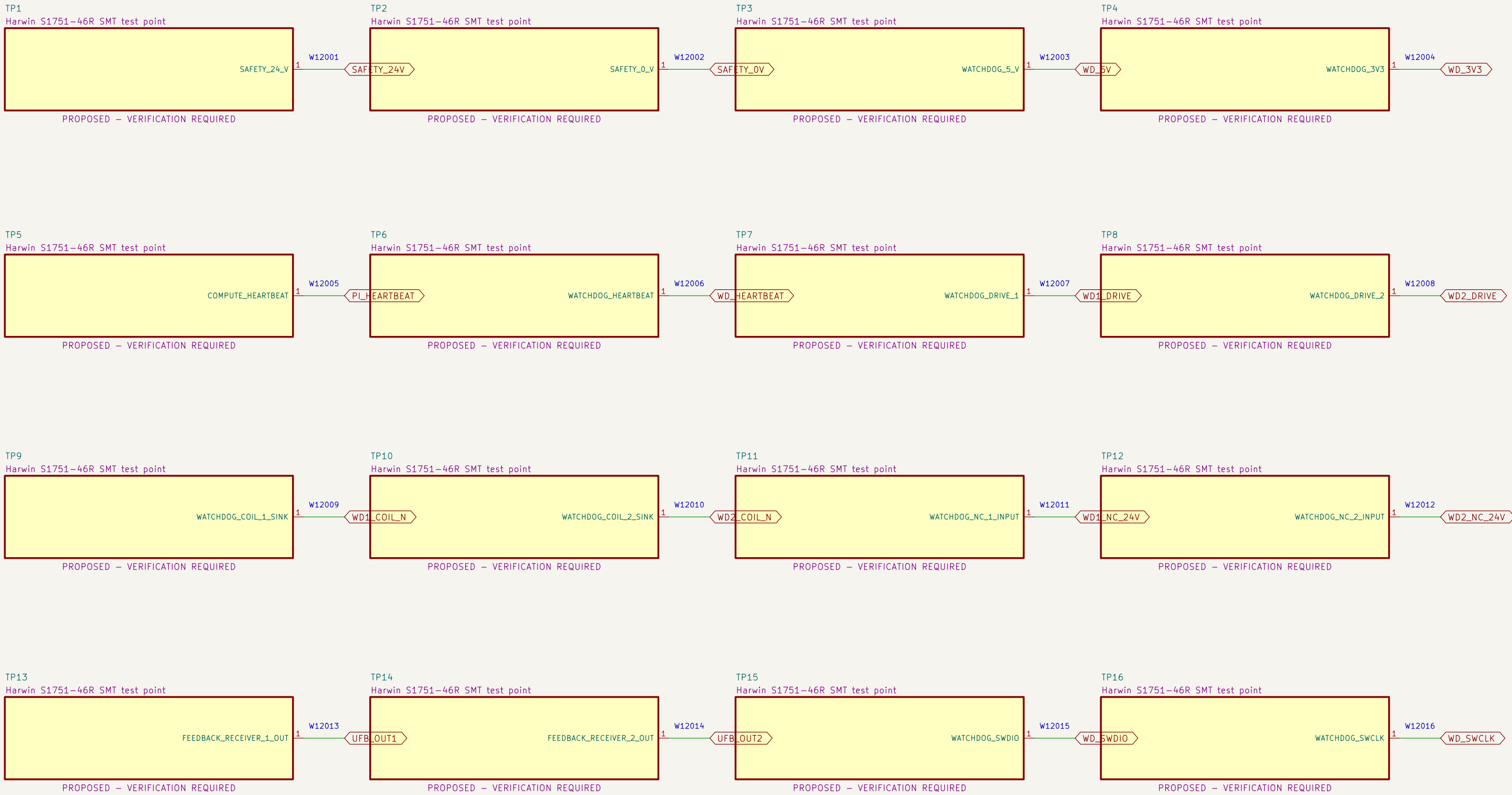
NOTE 1: Terminal numbering follows the PCB footprint and must be confirmed against received parts before wiring.

NOTE 2: Nominal terminal ratings do not release branch protection, conductor, ferrule, harness, enclosure or thermal application.

|                                                                                            |                  |               |
|--------------------------------------------------------------------------------------------|------------------|---------------|
| CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED                                     |                  |               |
| PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION                                 |                  |               |
| Project Button                                                                             |                  |               |
| Sheet: /11 Watchdog PCB external connectors/<br>File: 11_watchdog_pcb_connectors.kicad_sch |                  |               |
| Title: PB HR–V0 V3–P1.13 – 11                                                              |                  |               |
| Size: A3                                                                                   | Date: 2026–08–08 | Rev: V3–P1.13 |
| KiCad E.D.A. 10.0.5                                                                        |                  | Id: 12/13     |

12 Watchdog PCB test access

Exact clip-compatible test terminals expose controlled nodes without granting safety or operating authority.



NOTE 1: Test terminals are diagnostic features only; they provide no safety integrity or bypass authority.

NOTE 2: Verify clip clearance with guards, harnesses and power removed before any live measurement procedure is released.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED  
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION

Project Button

Sheet: /12 Watchdog PCB test access/  
File: 12\_watchdog\_pcb\_test\_access.kicad\_sch

Title: PB HR-V0 V3-P1.13 – 12

Size: A3

Date: 2026-08-08

Rev: V3-P1.13

KiCad E.D.A. 10.0.5

Id: 13/13